☑ Midterm Examination – SPRING 2023

Course Title:	Design and Analysis of Algorithms				Course Code:	CSC301	Credit Hours: 3(3,0)
Course Instructor:	Dr. Hasan Jamal				Programme Name:	BS Computer Science	
Semester:	5th	Batch:	SP21-BCS	Section:	A, B, C	Date:	05/05/2023
Time Allowed:	90 Minutes				Maximum Marks:	50	
Student's Name:	[redacted]				Reg. No.	[redacted]	

Important Instructions / Guidelines:

- Be precise and to the point while answering any question.
- Show all your work, as partial credits will be given. You will be graded not only on the correctness of your answer, but also on the clarity with which you express it. Please be neat.
- Good luck!

Question 1: CLO: <1>; Bloom Taxonomy Level: <Understanding> [Marks: 5]

What are the limitations of empirical analysis? Why do we need to theoretically analyze algorithms?

Question 2: CLO: <2>; Bloom Taxonomy Level: <Analyzing> [Marks: 10]

Prove or disprove: $n^2 - 2n - 5 \in \Theta(n^2)$

Question 3: CLO: <2>; Bloom Taxonomy Level: <Analyzing> [Marks: 10]

For the following code snippet, provide a line-by-line analysis and construct function T(n) that give the runtime of this code snippet as a function of "n". Also determine the Big-Oh of this code snippet.

```
x = n; -> 1.
while (x > 0) {(x+1) -> (n+1).
        if ( x < i)
            sum += foo( x ) * bar( x-1);
        else
            sum += bar( x );
        x = x - 1 -> 1.
}

bar (a) {
        for (int i = 0; i < n; ++i) => (n+1).
            for (int j = 0; j < i; ++j)   ->   (n+1).n.
                sum += a*(i + j);              n.n. => n^2 + n^2 + n+n+1 => 2n^2+2n+1
        return sum;
}

foo (a) {
        for (int i = 0; i < n*n; ++i)  n^2   n^2.2n^2+1.
            sum += a * i;
        return sum; -> 1.
}
```

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Question 4: CLO: <2>; Bloom Taxonomy Level: <Analyzing> [Marks: 10]

In not more than five lines each, provide the answers to the following questions. Be reasonable in your assumptions. Justify your answer.

a) You are given an unsorted list of 1,000 integers with repeating numbers ranging from 0 to 1,000. Which sorting algorithm will you choose to sort this data set? Justify your answer.

b) You have a uniformly distributed set of unsorted integers of 5,000 numbers, within the range 0 to 100,000. Which algorithm will you choose to sort this data set? Justify your answer.

c) You have a list of share prices of 500 stocks. To sort this list, you need a stable algorithm with an absolute guarantee of O(n lg n) performance in the worst case. Which algorithm will you choose? Justify your answer.

d) You have an unsorted list of final exam marks of a class. You would like to sort the data so that you can grade students on the curve. Which algorithm will you choose to sort this list? Justify your answer.

Question 5: CLO: <2>; Bloom Taxonomy Level: <Analyzing> [Marks: 10]

Solve the following recurrence using the "Recursion Tree Method"

$$T(n) = 2T\left(\frac{n}{4}\right) + 1$$

Question 6: CLO: <2>; Bloom Taxonomy Level: <Analyzing> [Marks: 5]

Solve the following recurrence using the "Master Theorem Method"

$$T(n) = 4T\left(\frac{n}{4}\right) + n$$

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